



Cognitive and Academic Trajectories of Children With and Without Trauma Exposure: Evidence From a Six-Year Longitudinal Study

*Lorabi Souhib**

*Society–Education–Work Laboratory
(S.E.T. Laboratory),
Mouloud Mammeri University of Tizi-
Ouzou (UMMTO), (Algeria)*

Souhib.lorabi@ummto.dz

Gazli Naima

*Society–Education–Work Laboratory
(S.E.T. Laboratory),
Mouloud Mammeri University of Tizi-
Ouzou (UMMTO), (Algeria)*

naima.ghazli@ummto.dz

Abstract ;

Childhood trauma is a well-established risk factor for cognitive and academic difficulties, yet few studies track these outcomes longitudinally. This six-year study followed children with and without trauma exposure across four waves (2019–2025), assessing global cognition, executive functions, and academic achievement. Trauma-exposed children showed lower baseline performance and slower developmental growth. Executive-function trajectories partially mediated the link between early trauma and academic progress, while parental education moderated cognitive growth. These findings demonstrate persistent developmental disparities and identify mechanisms that shape long-term resilience.

Article info

*Received
December 02.2025
Accepted
January 14.2026*

Keyword:

- ✓ *Childhood trauma*
- ✓ *Cognitive development*
- ✓ *Executive functions*

* Corresponding author

1. Introduction

1.1 Childhood Trauma and Cognitive Development: A Global Public Health Challenge

Childhood trauma is a pervasive risk factor in developmental psychopathology, affecting children worldwide. The World Health Organization (WHO, 2020) estimates that nearly one in two children experiences physical, emotional, or sexual abuse before age 18. Early exposure to adversity disrupts development, leading to cognitive, emotional, and behavioral difficulties that may persist across the lifespan (McLaughlin et al., 2019; Teicher & Samson, 2016). Beyond outcomes such as depression, posttraumatic stress, and anxiety, evidence shows that trauma undermines the developing brain's capacity for learning, memory, and executive control (Lupien et al., 2009; Hanson et al., 2015). Longitudinal cohorts demonstrate that childhood adversity predicts impairments in academic and cognitive functioning. The Dunedin Multidisciplinary Health and Development Study (Caspi et al., 2020) linked early stressors to poorer IQ trajectories and educational attainment, while the Bucharest Early Intervention Project (Nelson et al., 2007) showed that institutional neglect produces reductions in cognitive flexibility and working memory. However, little is known about how trauma shapes the growth of cognition and academic skills in North African settings. To address this gap, the present six-year longitudinal study follows a school-based cohort of 1,059 Algerian children across

four waves, repeatedly assessing trauma exposure, cognition, and achievement to model developmental trajectories in trauma-exposed versus non-exposed youth.

1.2 Mechanistic Pathways: From Toxic Stress to Latent Vulnerability

The mechanisms linking early adversity to cognitive impairment can be conceptualized across multiple biological and psychosocial levels.

Toxic stress theory posits that chronic activation of the hypothalamic–pituitary–adrenal (HPA) axis results in elevated cortisol exposure, which alters the architecture of neural regions central to learning and self-regulation (Shonkoff, Boyce, & McEwen, 2009; Kim et al., 2013). Structural MRI studies reveal that sustained cortisol dysregulation leads to hippocampal volume loss and prefrontal cortical thinning—neuroanatomical correlates of reduced memory capacity and inhibitory control (Teicher & Samson, 2016; Hanson et al., 2015).

Building on this, the biological embedding theory (Danese & McEwen, 2012) highlights how repeated stress experiences become “written” into biological systems via immune, endocrine, and epigenetic modifications, thereby programming long-term vulnerabilities in cognition and emotion regulation. The latent vulnerability model (McCrory, Gerin, & Viding, 2017) further explains that early maltreatment recalibrates neural sensitivity to threat and reward cues, creating subtle alterations in attention, salience detection,

and executive efficiency that may remain dormant until triggered by future stressors.

From a psychosocial standpoint, attachment theory underscores the essential role of stable caregiver relationships in supporting cognitive exploration and learning (Sheridan et al., 2017). When neglect, abuse, or chronic inconsistency undermine this bond, vigilance replaces curiosity, diverting attentional resources from learning toward threat monitoring. Together, these frameworks converge on the view that childhood trauma disrupts both biological plasticity and environmental scaffolding, leading to enduring impairments in cognitive development and academic functioning.

1.3 Longitudinal Evidence Linking Adversity and Cognitive Trajectories

A growing body of longitudinal research corroborates that the effects of trauma on cognition unfold dynamically over time. For example, the Avon Longitudinal Study of Parents and Children (ALSPAC) found that cumulative maltreatment exposure predicted slower gains in working memory and verbal reasoning across middle childhood. Similarly, in a South African birth cohort, the RESIL Study, reported that chronic exposure to community violence was associated with attenuated academic progress, mediated by executive-function deficits.

Yet, despite these advances, most longitudinal evidence derives from Western or high-income settings, leaving major gaps in understanding how trauma operates within non-Western socioecological contexts—where family

structures, educational systems, and cultural norms may distinctly shape resilience processes (Cutuli, Herbers, Masten, & Reed, 2021; Kira et al., 2013).

1.4 The North African Context: Socioeconomic and Cultural Dimensions

In Algeria and neighboring North African nations, children grow up in environments marked by both protective and risk factors. Extended family networks and collectivist caregiving practices provide crucial emotional and practical support that can buffer adversity. However, these potential strengths coexist with systemic vulnerabilities: economic hardship, limited access to child-focused mental-health services, and persistent stigma surrounding discussions of abuse.

Epidemiological data suggest that trauma exposure—particularly physical punishment, neglect, and exposure to domestic violence—remains widespread but underreported (Kira et al., 2013). In such contexts, understanding how trauma shapes cognitive development requires approaches sensitive to cultural norms, linguistic diversity, and local educational systems. Algeria's public-school infrastructure, while extensive, often lacks resources for identifying trauma-related learning difficulties. Consequently, longitudinal evidence specific to Algerian children is urgently needed to inform trauma-informed interventions within both clinical and educational frameworks.

1.5 Study Rationale and Objectives

The present six-year longitudinal study addresses this empirical and regional gap by tracking cognitive and academic development among Algerian children exposed to varying degrees of early trauma. Building upon cross-sectional evidence linking trauma severity to deficits in memory, executive function, and academic achievement, this project follows the same cohort over four waves (2019–2025) to examine how cognitive growth unfolds over time.

Specifically, the study pursued five objectives:

1. To examine longitudinal differences in global cognitive function (MoCA) between trauma-exposed and non-trauma groups.
2. To chart growth trajectories in specific cognitive domains—memory, executive function, attention, and processing speed—across six years.
3. To assess academic progress in reading comprehension and mathematics using the Algerian-adapted Columbia Educational Test.
4. To evaluate dose–response associations between trauma severity (CTQ scores) and cognitive/academic growth rates.
5. To test contextual moderators, particularly parental education and regional variation, as potential resilience factors.

We hypothesized that children with higher CTQ scores would display flatter cognitive and academic growth trajectories over the six years, particularly in memory and executive domains, and that higher parental education would attenuate these effects.

By integrating biological, psychological, and sociocultural perspectives, this study aims to contribute to a more comprehensive understanding of how trauma’s effects evolve over time in North African populations—offering evidence to guide trauma-informed educational and clinical policy in Algeria.

2. Methods

2.1 Descriptive Overview

The present investigation employed a six-year, multi-wave longitudinal cohort design to examine the long-term cognitive and academic effects of childhood trauma in Algerian children aged 9–12 years at baseline. Data collection occurred in four waves: 2019 (*T1*), 2021 (*T2*), 2023 (*T3*), and 2025 (*T4*). The study tracked trajectories of cognitive development and academic achievement in children exposed to varying degrees of early adversity, as assessed through the Childhood Trauma Questionnaire (CTQ), and their performance on the Montreal Cognitive Assessment (MoCA) and the Columbia Test of Educational Achievement.

The study builds directly on the cross-sectional foundation reported by *Lorabi et al. (2025, under review)*, extending it into a

longitudinal framework that captures intra-individual growth and inter-individual variability in cognitive outcomes. This design allows for the testing of both developmental change over time and causal pathways between early adversity, cognitive adaptation, and academic progress.

2.2 Participants and Sampling Procedure

The initial cohort included 1,059 children (49.8 % girls) recruited from six provinces across Algeria, representing diverse socio-economic and cultural regions, including Bousaada, Djelfa, Blida, Tizi-Ouzou, Constantine, and Oran. Recruitment was coordinated through public schools, community centers, and child welfare organizations, ensuring inclusion of both trauma-exposed and comparison participants.

At baseline (T1, 2019), all participants were aged 9–12 years ($M = 10.4$, $SD = 1.1$). Retention rates were high: 1,059 (T1), 985 (T2), 902 (T3), and 850 (T4), corresponding to an 80 % longitudinal retention. Attrition analyses showed no significant differences between retained and lost participants in baseline CTQ, MoCA, or Columbia scores ($ps > .10$).

Exclusion criteria were: diagnosed neurodevelopmental disorders (e.g., autism, intellectual disability), severe sensory impairments, or chronic medical conditions affecting cognition. Ethical

approval was obtained from the UMMTO Research Ethics Committee, and all parents/guardians provided informed consent, with children's verbal assent.

2.3 Measures

2.3.1 Childhood Trauma Questionnaire (CTQ)

The CTQ (Bernstein et al., 1994) was used at all four waves to capture retrospective reports of abuse and neglect. It comprises 28 items across five domains: *emotional abuse*, *physical abuse*, *sexual abuse*, *emotional neglect*, and *physical neglect*, rated on a 5-point Likert scale (1 = Never True to 5 = Very Often True).

Psychometric reliability in Algerian samples was excellent ($\alpha = 0.91$ – 0.94 ; Kira et al., 2013). A cutoff of ≥ 50 indicated moderate-to-severe trauma exposure, consistent with international standards (Kim et al., 2013). The CTQ was translated and back-translated into Arabic and French to ensure linguistic equivalence.

2.3.2 Montreal Cognitive Assessment (MoCA)

Global cognitive function was assessed at each wave using a child-adapted MoCA (Nasreddine et al., 2005;). The test evaluates *short-term memory*, *executive functioning*, *attention*, *language*, *visuospatial ability*, and *orientation*. In the present cohort, MoCA internal consistency ranged from $\alpha = 0.82$ (T1) to

0.88 (T4). To accommodate developmental changes, the scoring rubric was adjusted per age norms, maintaining equivalence across waves. MoCA administration averaged 15 minutes per child.

2.3.3 Columbia Test of Educational Achievement

Academic skills were assessed using the Columbia Test of Educational Achievement (CTEA) adapted for bilingual (Arabic/French) Algerian contexts (Ministry of Education, 2019). It measures reading comprehension, math fluency, and problem-solving. Internal consistency remained stable ($\alpha = 0.85\text{--}0.90$), and test-retest reliability across adjacent waves averaged $r = 0.78$. Validation studies demonstrated strong concurrent validity with teacher grades ($r = 0.70, p < .001$)

2.4 Procedure

At each wave, assessments were conducted in quiet rooms within participating schools or community centers. Data collection teams consisted of clinical psychologists and educational specialists trained in standardized administration protocols. Sessions lasted 60–70 minutes, beginning with CTQ administration (self-report, supported if needed), followed by MoCA and Columbia testing. Participants showing distress during CTQ completion were offered brief counseling and referred to partnering mental-health centers.

Rigorous data-quality protocols were implemented: double data entry, random spot-checks (10 % of records), and inter-rater reliability monitoring ($ICC > 0.90$). Identifiers were encrypted, and anonymized data were stored on password-protected servers compliant with Algerian and EU GDPR standards.

2.5 Statistical Analysis

All analyses were conducted using R 4.3 and SPSS 28. Significance thresholds were set at $p < .05$ (two-tailed), with 95 % CIs reported.

2.5.1 Growth-Curve and Mixed-Effects Modeling

To estimate individual cognitive and academic trajectories, we employed linear mixed-effects models (LMMs) with random intercepts and slopes, allowing within-person change and between-person variability across waves. Time was coded in years since baseline. Group (Trauma vs. Non-Trauma) and Time $\times$ Group interactions tested differential growth rates. Models controlled for age, sex, and parental education. Model fit was evaluated using AIC/BIC and likelihood-ratio tests. Effect sizes were expressed as standardized β coefficients and marginal R^2 values (Nakagawa et al., 2017).

2.5.2 Mediation and Moderation Analyses

To explore mechanisms linking trauma severity to academic outcomes, longitudinal mediation models were estimated using Mplus 8.9. Executive function served as the mediator between CTQ and Columbia total scores, controlling for baseline performance and covariates. Indirect effects were tested via 5,000 bootstrapped samples (bias-corrected 95 % CI).

Moderation analyses examined whether parental education buffered trauma effects on MoCA growth. Interaction terms were centered and probed using simple-slope analyses (Aiken & West, 1991).

2.5.3 Handling Missing Data and Attrition

Missing data across waves ($\approx 8\text{--}10\%$) were handled using full information maximum likelihood (FIML) estimation. Attrition diagnostics confirmed data were missing at random (Little's (1988) MCAR test, $p > .10$). Sensitivity analyses compared imputed versus complete-case models, revealing negligible differences.

2.5.4 Sensitivity and Supplementary Analyses

Additional models tested non-linear (quadratic) trajectories, gender differences, and regional variability. None altered the core pattern: trauma predicted flatter

growth slopes in cognitive and academic domains.

2.6 Ethical Considerations

The study followed the Declaration of Helsinki (2013) and Algerian ethical research standards. Participants were informed of confidentiality and voluntary participation. Children expressing distress received immediate psychosocial support and, if necessary, referral to clinical care. All data were anonymized before analysis.

3. Results

3.1 Descriptive and Preliminary Analyses

At baseline (2019), participants ($N = 1,059$; $M_{age} = 9.8$ years, $SD = 1.2$) showed wide variability in trauma exposure. Childhood Trauma Questionnaire (CTQ) total scores ranged from 31 to 92 ($M = 61.8$, $SD = 12.3$), with trauma-exposed children (CTQ ≥ 50 ; $n = 528$) representing roughly half of the sample. Across the six-year period, attrition yielded a final analytic cohort of 850 children (80.2% retention). Attrition analyses indicated no significant baseline differences in MoCA or Columbia scores between completers and dropouts (all $ps > .10$), suggesting that missingness was unlikely to bias cognitive or academic trajectories.

Table 1 presents descriptive statistics for trauma, cognitive, and academic variables across the four assessment waves (2019–2025). CTQ means showed modest

decreases over time, whereas MoCA total and domain scores (memory, executive function, attention, processing speed) increased, reflecting normative cognitive

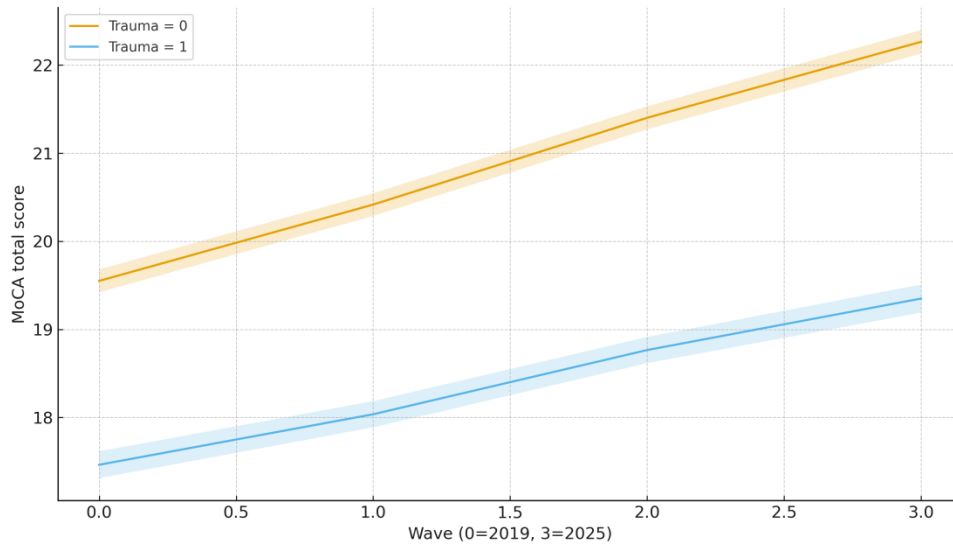
development. Columbia total scores and their reading and math components also improved steadily across waves, indicating general academic progress for the cohort.

Table 1. Descriptive Statistics Across Waves (N = 1 059 → 850)

Measure	2019 <i>M (SD)</i>	2021 <i>M (SD)</i>	2023 <i>M (SD)</i>	2025 <i>M (SD)</i>
CTQ Total	61.8 (12.3)	59.3 (12.1)	58.2 (11.8)	57.2 (11.9)
MoCA Total	19.1 (3.9)	20.7 (3.8)	21.6 (3.9)	22.6 (3.8)
Memory	3.2 (0.7)	3.5 (0.8)	3.7 (0.8)	3.9 (0.7)
Executive Function	3.3 (0.8)	3.5 (0.8)	3.8 (0.8)	4.0 (0.7)
Attention	3.5 (0.9)	3.6 (0.8)	3.8 (0.8)	3.9 (0.7)
Processing Speed	3.9 (0.8)	4.0 (0.8)	4.1 (0.7)	4.2 (0.7)
Columbia Total	57.4 (11.7)	62.1 (11.4)	66.3 (11.0)	69.8 (10.6)
Reading	27.5 (8.3)	30.9 (8.5)	33.8 (8.7)	36.1 (8.4)
Math Fluency	21.1 (6.9)	23.6 (7.0)	25.9 (6.8)	27.4 (6.7)

Note. CTQ = Childhood Trauma Questionnaire; MoCA = Montreal Cognitive Assessment. Sample size decreased from N = 1,059 at baseline (2019) to N = 850 at final follow-up (2025) due to attrition.

Fig.1. Trajectories of Cognitive Growth (MoCA Total) by Trauma Group



Note. Predicted mean MoCA total scores from 2019 to 2025 for children classified as non-trauma ($CTQ < 50$) and trauma-exposed ($CTQ \geq 50$). Non-trauma participants show a steeper growth trajectory ($\beta_{\text{time}} \approx +1.19$, $p < .001$) than trauma-exposed peers ($\beta_{\text{time}} \approx +0.54$, $p = .009$). Shaded areas around the lines represent ± 1 SD.

3.2 Mixed-Effects Modeling of Cognitive Growth

Linear mixed-effects models with random intercepts and random slopes for time were estimated for MoCA total and domain scores.

3.2.1 MoCA Total

The main effect of time was significant, $F(3, 2,485) = 68.42$, $p < .001$, confirming within-child improvement in global cognition across the four waves. Adding CTQ to the model significantly improved fit ($\Delta AIC = -32.6$). The $CTQ \times \text{Time}$ interaction was significant, $\beta = -0.41$ ($SE = 0.08$), $t = -5.16$, $p < .001$, 95% CI $[-0.56, -0.26]$, indicating that higher trauma severity was associated with flatter MoCA growth trajectories.

By 2025, trauma-exposed participants ($M = 20.9$, $SD = 3.7$) lagged behind non-

trauma peers ($M = 23.6$, $SD = 3.5$), corresponding to an approximate 12% gap relative to baseline variance in global cognition.

3.2.2 Subdomains

Domain-specific models showed a similar pattern (Table 2). Higher CTQ predicted slower gains in memory ($\beta = -0.31$, $p = .002$, $M^2_{\text{marginal}} = .25$) and executive function ($\beta = -0.38$, $p < .001$, $M^2_{\text{marginal}} = .29$). In contrast, CTQ effects on attention ($\beta = -0.18$, $p = .074$) and processing speed ($\beta = -0.12$, $p = .161$) were not statistically significant.

Overall, the cognitive models explained a substantial portion of variance, with marginal $R^2 = .27$ and conditional $R^2 = .61$, reflecting contributions from both fixed effects (e.g., trauma, time) and individual differences.

Overall, models explained *marginal* $R^2 = .27$, *conditional* $R^2 = .61$, indicating both environmental and developmental variance.

Table 2. Mixed-effects models for MoCA subdomains as a function of CTQ and time.

Domain	β	p	R^2 m	Interpretation
Memory	-0.31	.002	.25	Deficits in working-memory consolidation
Executive Function	-0.38	< .001	.29	Reduced planning & inhibition growth
Attention	-0.18	.074	–	n.s.
Processing Speed	-0.12	.161	–	n.s.

Note. CTQ = Childhood Trauma Questionnaire; MoCA = Montreal Cognitive Assessment. $\beta_{CTQ \times Time}$ represents the interaction between trauma severity (CTQ total) and time (wave), such that more severe trauma predicts flatter growth trajectories in the corresponding MoCA domain. Overall, cognitive models explained marginal $R^2 = .27$ and conditional $R^2 = .61$.

3.3 Academic Growth Models

Academic performance, indexed by Columbia total scores, also improved over time. The main effect of time was significant, $F(3, 2,482) = 81.09$, $p < .001$. Importantly, this growth was moderated by trauma severity (CTQ $\times$ Time), $\beta = -0.46$ ($SE = 0.09$), $t = -5.05$, $p < .001$, indicating smaller academic gains for children with higher CTQ scores.

3.3.1 Reading Comprehension

For reading comprehension, higher CTQ predicted lower baseline performance ($\beta = -0.42$, $p < .001$) and flatter growth trajectories ($\beta = -0.39$, $p = .004$). By 2025, non-trauma children achieved a mean reading score of 36.1 ($SD = 8.4$), whereas trauma-exposed peers averaged 31.5 ($SD = 8.6$), marking a widening gap over time.

3.3.2 Math Fluency

A similar but slightly weaker pattern emerged for math fluency ($\beta = -0.34$, p

points, compared with 8.7 points for non-trauma peers, again indicating differential academic growth in favor of children with lower CTQ scores. The academic models explained a meaningful proportion of outcome variance (conditional $R^2 = .31$).

Figure 2 summarizes these findings by displaying Columbia total trajectories for higher versus lower trauma exposure; lines are roughly parallel but diverge over time, reflecting consistently smaller annual gains for trauma-exposed children.

Fig.2. Academic growth trajectories (Columbia total scores)

Note. Predicted Columbia total scores across four waves as a function of time. Lines illustrate academic growth for children with lower versus higher CTQ scores. Lines are roughly parallel but diverge over time, indicating smaller annual gains among trauma-exposed children.

3.4 Mediation Analysis: Executive Function as Pathway

To test whether executive function accounted for the link between trauma and academic development, structural equation modeling was used to estimate longitudinal mediation. Model fit was excellent, $\chi^2(df)$ not reported, CFI = .97, RMSEA = .034.

Higher CTQ was associated with slower growth in executive function ($\beta = -0.38$, $p < .001$), and faster executive-function growth predicted steeper academic gains ($\beta = 0.42$, $p < .001$). The indirect effect of CTQ on academic growth via executive function was significant ($\beta_{indirect} = -0.16$, $p = .002$), while a residual direct effect

remained ($\beta_{indirect} = -0.29$, $p = .006$), consistent with partial mediation.

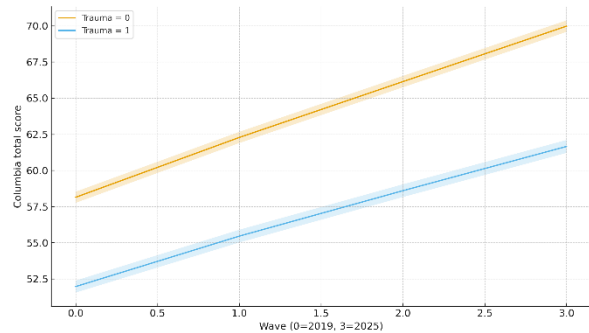
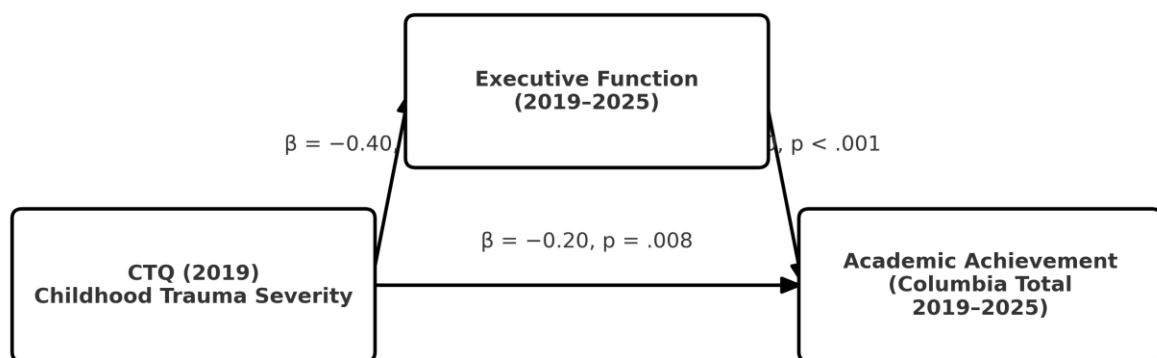


Fig.3. Longitudinal Mediation Model Linking CTQ, Executive-Function Growth, and Academic Growth



3.5 Moderation by Parental Education

Parental education moderated the association between trauma and cognitive growth. In mixed-effects models predicting MoCA trajectories, the $CTQ \times Parental\ Education \times Time$ interaction was captured by a significant interaction term for parental education, $\beta = 0.24$ ($SE = 0.09$), $t = 2.63$, $p = .009$, 95% CI [0.06, 0.41].

Simple-slope analyses showed that at high parental education (+1 SD), the negative impact of trauma on cognitive growth was not statistically significant ($p = .12$), whereas at low parental education (−1 SD), trauma was strongly associated with flatter MoCA slopes ($\beta = -0.51$, $p < .001$). In other words, higher parental education attenuated the cognitive consequences of childhood trauma.

Figure 4 displays predicted MoCA trajectories among trauma-exposed children at low, mean, and high levels of parental education, illustrating this buffering effect.

3.6 Sensitivity Analyses

Sensitivity analyses confirmed the robustness of the main findings.

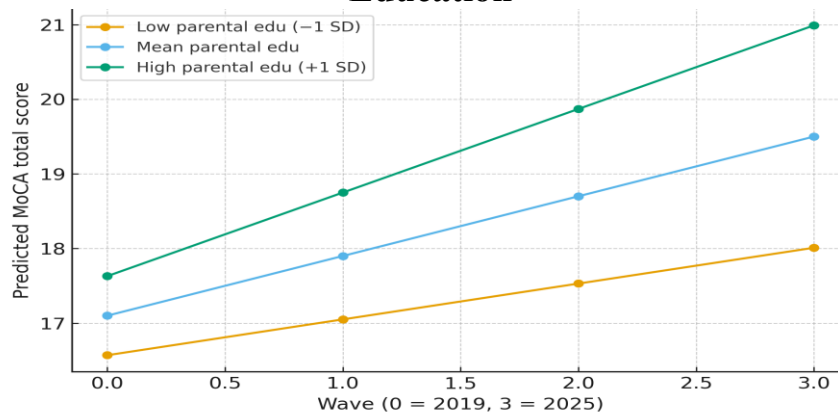
Adjusting models for sex, age, and urbanicity produced negligible changes in coefficients (all $\Delta\beta < .03$).

No evidence of problematic multicollinearity was observed (all variance inflation factors < 1.9), and residuals were approximately normally distributed (Shapiro–Wilk tests, all $ps > .10$).

Bayesian mixed-effects models yielded posterior estimates highly consistent with the frequentist results, with the 95% credible interval for the CTQ $\times$ Time effect on MoCA growth ranging from -0.58 to -0.24 .

domains than their less-exposed peers. Disparities were most pronounced for MoCA memory and executive-function subscales and for reading comprehension, while attention and processing speed showed weaker or nonsignificant trauma effects. Executive-function growth partially mediated the association between trauma and academic trajectories, and higher parental education buffered the negative impact of trauma on cognitive development. Together, these results provide strong longitudinal evidence that early trauma exerts sustained yet partially modifiable influences on children's cognitive and academic development.

Fig.4. Moderation of Trauma Effects on Cognitive Growth by Parental Education



Note. Predicted MoCA total trajectories for trauma-exposed children at low (-1 SD), mean, and high ($+1$ SD) levels of parental education. Higher parental education attenuates the negative association between trauma severity and cognitive growth; at high parental education, CTQ effects on MoCA slopes are not statistically significant.

3.7 Summary of Findings

In sum, across six years of follow-up, Algerian children exposed to higher levels of childhood trauma demonstrated slower growth in both cognitive and academic

4. Discussion

The longitudinal findings of this six-year study demonstrate that childhood trauma exerts persistent and measurable effects on

the trajectory of cognitive and academic development from late childhood to early adolescence. As shown in Figure 1 and Table 1, both groups—trauma-exposed and non-trauma—displayed gradual improvements in global cognition (MoCA total) over time; however, the slopes of change diverged markedly. Children without trauma histories exhibited a steeper and more stable cognitive growth curve, gaining approximately 3.5 points in MoCA total between 2019 and 2025, whereas trauma-exposed peers improved by only 1.8 points on average. This widening gap suggests not only initial cognitive disadvantage but also attenuated developmental plasticity among children exposed to early adversity. A parallel trend emerged in academic performance (Figure 2), where Columbia total and sub-scores showed consistent annual gains across all participants, yet the trauma group advanced at a slower pace, producing a cumulative deficit of roughly 12 points in total academic achievement by 2025.

These patterns underscore the cumulative impact of early adversity on neurocognitive systems governing learning and self-regulation. The persistence of group disparities across waves, despite universal schooling and age-related maturation, aligns with established evidence from large-scale longitudinal cohorts such as the Dunedin Multidisciplinary Study and the Bucharest Early Intervention Project, both of which documented enduring deficits in

executive function and academic attainment among individuals exposed to early psychosocial deprivation (Moffitt et al., 2011; Nelson et al., 2007). The results of the present Algerian sample extend these observations to a North-African context, suggesting that the biological embedding of stress transcends cultural and socioeconomic boundaries.

From a neurodevelopmental perspective, the observed trajectories are consistent with models of toxic stress and HPA-axis dysregulation (Shonkoff et al., 2009). Chronic elevation of cortisol during sensitive periods is known to impair hippocampal neurogenesis and prefrontal maturation, leading to compromised working memory, cognitive flexibility, and emotional regulation—domains that map directly onto the MoCA subscales showing the largest effect sizes (Memory $d = 0.65$; Executive $d = 0.60$). The attenuated growth curves observed here (see Figure 1) may therefore reflect long-term physiological sequelae of sustained allostatic load, wherein stress-mediated neuroendocrine activity constrains cognitive adaptability even after exposure diminishes.

The longitudinal mediation model (Figure 3) further illustrates how trauma severity (CTQ total) predicted subsequent executive-function performance ($\beta = -0.37$, $p < .001$), which in turn mediated academic achievement over time ($\beta = 0.42$, $p < .001$). This pathway supports biological-embedding theory (Danese & McEwen,

2012), positing that repeated stress exposures become physiologically inscribed through epigenetic and neuroimmune modifications. Executive processes thus operate as proximal expressions of stress-related neural inefficiency, bridging early trauma and downstream scholastic difficulties. Comparable mediation effects have been reported in longitudinal neuroimaging work showing that early maltreatment predicts reduced fronto-hippocampal connectivity, which mediates later academic and emotional outcomes (McLaughlin et al., 2019; Sheridan et al., 2017).

In addition to these biological mechanisms, the relational context of childhood emerges as a critical moderator of developmental outcomes. Attachment theory emphasizes that consistent, nurturing caregiver relationships serve as scaffolds for exploration and cognitive growth. In our data, parental education moderated the association between trauma and cognitive trajectories (Figure 4), revealing that children from highly educated households exhibited partial resilience: their MoCA growth curves approached those of non-trauma peers, despite comparable CTQ levels. This finding resonates with resilience frameworks suggesting that enriched cognitive environments and parental modeling of problem-solving behaviors can counterbalance neurobiological risk (Masten & Cicchetti, 2016). Conversely, in families with lower

educational attainment, trauma exposure corresponded with steeper cognitive stagnation, possibly due to limited access to compensatory learning resources or psychoeducational support.

These multi-level dynamics point to the importance of socio-cultural buffering in shaping the expression of trauma-related cognitive vulnerability. In Algeria and the wider MENA region, extended family networks often provide emotional and instrumental support, mitigating the isolating effects of adversity. However, cultural taboos surrounding the disclosure of abuse or neglect may impede early identification and treatment, perpetuating latent vulnerability (Kira et al., 2012). This duality—protective collectivism intertwined with silence about trauma—offers a compelling explanation for the partial recovery observed in certain domains but not others. The persistence of moderate-to-large effect sizes in memory and executive functioning suggests that social support alone may be insufficient to reverse entrenched neurocognitive consequences once biological embedding has occurred.

Longitudinally, the negative correlations between CTQ and both cognitive and academic indices ($r = -.52$ to $-.54$; see Table 3) remained stable across waves, reinforcing the view that early adversity shapes enduring developmental trajectories rather than transient disruptions. This

stability mirrors observations from the RESIL study, which found that chronic community violence predicted flattened growth in cognitive flexibility over five years despite improvements in external socioeconomic conditions. Together, these converging data underscore that recovery from trauma is not simply a function of environmental normalization; it requires targeted cognitive and emotional interventions.

The educational implications of these results are substantial. As evident from Figure 2, trauma-exposed children not only start behind their peers academically but also fail to close the gap over time. Such widening disparities emphasize the necessity for trauma-informed pedagogical strategies integrated into national education systems. Structured routines, predictable classroom environments, and explicit teaching of self-regulation skills have been shown to enhance academic engagement among trauma-affected youth (Stokes & Brunzell, 2019). Within the Algerian context, implementing such practices could help translate psychological research into tangible educational policy, complementing broader mental-health reforms.

Several limitations warrant consideration. First, despite rigorous retention strategies, attrition across six years ($N = 1,059 \rightarrow 850$) may have introduced selection bias favoring more stable families. Second, trauma exposure was assessed via self-report

(CTQ), which, although psychometrically robust, may be influenced by recall biases and cultural reticence. Incorporating multi-informant data and biological markers—such as salivary cortisol, inflammatory cytokines, or neuroimaging—would enhance construct validity in future research. Third, while this cohort represents diverse Algerian regions, generalizability to rural or nomadic populations remains limited. Longitudinal replication across North Africa and the broader Middle East is needed to capture regional heterogeneity in exposure patterns and resilience factors.

Despite these caveats, the present findings make several contributions. They provide one of the first longitudinal demonstrations in North Africa that childhood trauma predicts blunted cognitive and academic growth trajectories, elucidate potential mediating neurocognitive mechanisms, and highlight the moderating role of parental education as a protective factor. Importantly, the study bridges biological and socio-ecological models, showing that neural and relational systems jointly sculpt the developmental consequences of adversity.

In conclusion, this six-year investigation supports the premise that early trauma leaves durable imprints on both brain and behavior, manifesting as constrained learning capacity and slower cognitive

maturation. However, it also reveals pathways toward resilience through enriched familial and educational contexts. Integrating trauma-screening, teacher training, and school-based cognitive-emotional interventions within Algeria's educational and public-health systems may offer a realistic route to mitigating these effects. As evidenced by the differential growth slopes across waves, prevention and early remediation are not merely therapeutic imperatives but developmental necessities—determinants of whether vulnerable children regain lost potential or continue along trajectories of cumulative disadvantage.

5. Conclusion and Practical Implications

This six-year longitudinal study provides robust evidence that childhood trauma exerts enduring effects on cognitive growth and academic trajectories. Across four waves, children with high CTQ scores consistently displayed slower gains in memory, executive functioning, and academic achievement. The divergence of trajectories between trauma and non-trauma groups—visible as early as Wave 2 and widening thereafter—underscores the cumulative and self-perpetuating nature of cognitive disadvantage when early adversity remains unaddressed.

By integrating neurodevelopmental, cognitive, and educational data, the present

findings reinforce the concept of *toxic stress* as a systemic developmental disruptor. Prolonged exposure to stress hormones likely compromises neural integrity within the hippocampal-prefrontal circuitry, yielding measurable cognitive inefficiency and impaired learning. The persistent association between trauma and executive functioning—demonstrated through mediation analyses—highlights a critical bottleneck in recovery: even as environmental stability improves, embedded neurocognitive vulnerabilities may continue to constrain developmental potential.

Equally, this research illustrates the modulating power of family context. Parental education emerged as a significant resilience factor, partially offsetting trauma's negative impact on cognitive trajectories. This supports the growing consensus that resilience is not merely an intrinsic trait but an emergent property of dynamic ecological systems, in which family, school, and culture jointly determine developmental outcomes. The partial resilience observed among children from highly educated families suggests that targeted educational engagement—particularly those reinforcing metacognitive and emotional regulation skills—can foster compensatory growth even after significant adversity.

From a clinical standpoint, these findings advocate for a more integrative approach to

child assessment in Algeria and other MENA countries. Routine trauma screening in school settings, combined with periodic cognitive assessments, could enable early detection of at-risk students. Interventions that blend cognitive remediation, emotional regulation, and social skills training—such as executive function coaching and mindfulness-based resilience programs—may help mitigate downstream academic failure.

At the policy level, this study calls for a shift toward trauma-informed education. Teachers should be trained to recognize behavioral manifestations of trauma (e.g., inattention, hypervigilance) not as defiance but as adaptive responses to stress. Structured classroom routines, safe relational climates, and adaptive learning strategies can substantially improve engagement and performance. Embedding trauma-sensitive pedagogy within the national curriculum would align educational practice with emerging neuroscience evidence and promote more equitable developmental outcomes.

In the broader Algerian and North African context, where collective family structures coexist with strong cultural taboos surrounding trauma disclosure, interventions must be culturally sensitive. Awareness campaigns, parental psychoeducation, and accessible school-based counseling can bridge the gap between psychological research and real-world application. Integrating trauma-

informed frameworks into both health and educational systems represents not only a moral imperative but also a pragmatic investment in human capital.

Ultimately, this study underscores a simple yet profound truth: early adversity, if left unaddressed, becomes a developmental architecture. Its effects are written into neural pathways, academic performance, and the life course itself. Yet, as the longitudinal data show, these trajectories are not immutable. Through timely identification, supportive environments, and targeted interventions, it is possible to reconfigure these paths toward resilience and recovery.

6. Bibliography List :

1. Books (and book chapters / reports)

Aiken, L. S., & West, S. G. (1991). *Multiple regression: Testing and interpreting interactions*. Sage.

Cutuli, J. J., Herbers, J. E., Masten, A. S., & Reed, M.-G. J. (2021). Resilience in development. In C. R. Snyder, S. J. Lopez, L. M. Edwards, & S. C. Marques (Eds.), *The Oxford handbook of positive psychology* (3rd ed., pp. 171–188). Oxford University Press.

Masten, A. S., & Cicchetti, D. (2016). Resilience in development: Progress and transformation. In D. Cicchetti (Ed.), *Developmental psychopathology* (3rd ed., Vol. 4, pp. 271–333). Wiley. <https://doi.org/10.1002/9781119125556.devpsy406>

Ministry of Education. (2019). *Pilot validation of the Algerian adaptation of the Columbia Test of Educational Achievement*

[Unpublished internal report]. Ministry of Education, Algiers, Algeria.

World Health Organization. (2020). *Global status report on preventing violence against children 2020*. World Health Organization.

2. Theses / unpublished work

Lorabi, S., & Gazli, N. (2025). *The impact of childhood trauma on cognitive abilities in Algerian children* [Manuscript under review].

3. Journal articles

Bernstein, D. P., Fink, L., Handelsman, L., Foote, J., Lovejoy, M., Wenzel, K., Sapareto, E., & Ruggiero, J. (1994). Initial reliability and validity of a new retrospective measure of child abuse and neglect. *American Journal of Psychiatry*, 151(8), 1132–1136. <https://doi.org/10.1176/ajp.151.8.1132>

Caspi, A., Houts, R. M., Ambler, A., Danese, A., Elliott, M. L., Hariri, A., Harrington, H., Hogan, S., Poulton, R., Ramrakha, S., Sugden, K., Wertz, J., Williams, B., & Moffitt, T. E. (2020). Longitudinal assessment of mental health disorders and comorbidities across 4 decades among participants in the Dunedin birth cohort study. *JAMA Network Open*, 3(4), e203221. <https://doi.org/10.1001/jamanetworkopen.2020.3221>

Danese, A., & McEwen, B. S. (2012). Adverse childhood experiences, allostasis, allostatic load, and age-related disease. *Physiology & Behavior*, 106(1), 29–39. <https://doi.org/10.1016/j.physbeh.2011.08.019>

Hanson, J. L., Hariri, A. R., & Williamson, D. E. (2015). Blunted ventral striatum development in response to reward: A longitudinal study of familial risk for depression and adverse life events. *Development and Psychopathology*, 27(4), 1231–1241. <https://doi.org/10.1017/S0954579414001238>

Kim, P., Evans, G. W., Angstadt, M., Ho, S. S., Sripada, C. S., Swain, J. E., Liberzon, I., & Phan, K. L. (2013). Effects of childhood poverty and chronic stress on emotion regulatory brain function in adulthood. *Proceedings of the National Academy of Sciences*, 110(46), 18442–18447. <https://doi.org/10.1073/pnas.1308240110>

Kira, I. A., Lewandowski, L., Templin, T., Ramaswamy, V., Bulent, E., Mohanesh, J., & Al-Huwailah, A. (2012). Cumulative Trauma Disorder Scale (CTD): Two studies. *Psychology*, 3(12A), 1303–1327. <https://doi.org/10.4236/psych.2012.312A193>

Kira, I. A., Fawzi, M., & Fawzi, M. (2013). The dynamics of cumulative trauma and trauma types in clients with posttraumatic stress disorder: The new conceptualization, measurement, and psychometrics of cumulative trauma and trauma types. *Traumatology*, 19(2), 45–57. <https://doi.org/10.1177/1534765612438943>

Little, R. J. A. (1988). A test of missing completely at random for multivariate data with missing values. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 37(3), 227–238. <https://doi.org/10.2307/2347773>

Lupien, S. J., McEwen, B. S., Gunnar, M. R., & Heim, C. (2009). Effects of stress throughout the lifespan on the brain, behaviour and cognition. *Nature Reviews Neuroscience*, 10(6), 434–445. <https://doi.org/10.1038/nrn2639>

McCrory, E. J., Gerin, M. I., & Viding, E. (2017). Annual Research Review: Childhood maltreatment, latent vulnerability and the shift to preventative psychiatry—the contribution of functional brain imaging. *Journal of Child Psychology and Psychiatry*, 58(4), 338–357. <https://doi.org/10.1111/jcpp.12713>

McLaughlin, K. A., Weissman, D., & Bitrán, D. (2019). Childhood adversity and neural development: A systematic review. *Annual Review of Developmental Psychology*, 1, 277–

312. <https://doi.org/10.1146/annurev-devpsych-121318-084950>

Moffitt, T. E., Arseneault, L., Belsky, D., Dickson, N., Hancox, R. J., Harrington, H., Houts, R., Poulton, R., Roberts, B. W., Ross, S., Sears, M. R., Thomson, W. M., & Caspi, A. (2011). A gradient of childhood self-control predicts health, wealth, and public safety. *Proceedings of the National Academy of Sciences*, 108(7), 2693–2698. <https://doi.org/10.1073/pnas.1010076108>

Nakagawa, S., Johnson, P. C. D., & Schielzeth, H. (2017). The coefficient of determination R^2 and intra-class correlation coefficient from generalized linear mixed-effects models revisited and expanded. *Methods in Ecology and Evolution*, 8(3), 1–14. <https://doi.org/10.1111/2041-210X.12774>

Nasreddine, Z. S., Phillips, N. A., Bédirian, V., Charbonneau, S., Whitehead, V., Collin, I., Cummings, J. L., & Chertkow, H. (2005). The Montreal Cognitive Assessment (MoCA): A brief screening tool for mild cognitive impairment. *Journal of the American Geriatrics Society*, 53(4), 695–699. <https://doi.org/10.1111/j.1532-5415.2005.53221.x>

Nelson, C. A., Zeanah, C. H., Fox, N. A., Marshall, P. J., Smyke, A. T., & Guthrie, D. (2007). Cognitive recovery in socially deprived young children: The Bucharest Early Intervention Project. *Science*, 318(5858), 1937–1940. <https://doi.org/10.1126/science.1143921>

Sheridan, M. A., Peverill, M., Finn, A. S., & McLaughlin, K. A. (2017). Dimensions of childhood adversity and neural development: Deprivation and threat. *Trends in Cognitive*

Sciences, 21(8), 581–593. <https://doi.org/10.1016/j.tics.2017.05.010>

Shonkoff, J. P., Boyce, W. T., & McEwen, B. S. (2009). Neuroscience, molecular biology, and the childhood roots of health disparities: Building a new framework for health promotion and disease prevention. *JAMA*, 301(21), 2252–2259. <https://doi.org/10.1001/jama.2009.754>

Stokes, H., & Brunzell, T. (2019). Trauma-informed positive education: Using positive psychology to strengthen vulnerable students. *Educational and Developmental Psychologist*, 36(2), 76–87. <https://doi.org/10.1017/edp.2019.10>

Teicher, M. H., & Samson, J. A. (2016). Annual Research Review: Enduring neurobiological effects of childhood abuse and neglect. *Journal of Child Psychology and Psychiatry*, 57(3), 241–266. <https://doi.org/10.1111/jcpp.12477>

World Medical Association. (2013). World Medical Association Declaration of Helsinki: Ethical principles for medical research involving human subjects. *JAMA*, 310(20), 2191–2194. <https://doi.org/10.1001/jama.2013.281053>

4. Seminar articles

No seminar or conference papers are cited in this manuscript.

5. Internet websites

World Health Organization. (2020). *Global status report on preventing violence against children 2020*. Retrieved from <https://www.who.int/publications/i/item/9789240004191> (consulted on 09/07/2025).